TARGET BASED DRUG DISCOVERY FOR TYPE II DIABETES

AN INTERNSHIP REPORT

Submitted by

DEVADHARSHINI S

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

BIOTECHNOLOGY

DEPARTMENT OF BIOTECHNOLOGY

UNIVERSITY COLLEGE OF ENGINEERING

BHARATHIDASAN INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY, TIRUCHIRAPPALLI – 620 024

MAY 2025

UNIVERSITY COLLEGE OF ENGINEERING
BHARATHIDASAN INSTITUTE OF TECHNOLOGY
ANNA UNIVERSITY, TIRUCHIRAPPALLI – 620024

BONAFIDE CERTIFICATE

Certified that this project report **"TARGET BASED DRUG DISCOVERY FOR TYPE II DIABETES"** is the bonafide work of **"DEVADHARSHINI S (Reg no: 810021214012)"** who carried out the project work under my supervision**.**

SIGNATURE
Dr. P. SURESH KUMAR
HEAD OF THE DEPARTMENT
Professor
Department of Biotechnology
University College of Engineering
BIT Campus, Anna University
Tiruchirappalli – 620 024

SIGNATURE
Dr. V.TAMIL ELAKKIYA
MENTOR
Assistant Professor
Department of Biotechnology
University College of Engineering
BIT Campus, Anna University
Tiruchirappalli – 620 024

ACKNOWLEDGEMENT

It is with a profound feeling of fulfilment and gratitude, we are pleased to offer our most humble and heartfelt thanks to The Almighty for his enriching and encouraging presence all along the way in this arduous, but challenging scholastic pursuit.

We owe our sincere debt of gratitude to **Dr. P. SURESH KUMAR, Professor and Head, Department of Biotechnology, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli** for his abiding faith in our capabilities, optimistic encouragement and support in our academic endeavour.

We gratefully appreciate and acknowledge the valuable guidance and suggestions given by **Dr. V.TAMIL ELAKKIYA, Assistant Professor, Department of Biotechnology, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli**. We offer him our abundant gratitude, for the inspiring help and direction which was greatly instrumental, in the completion of our project.

We offer our grateful appreciation to the lab assistants and non-teaching staffs for their kind help and meaningful cooperation. Our heartfelt thanks to all my friends and seniors for their moral support and help throughout my project.

DEVADHARSHINI S

LIST OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	**VII**
	LIST OF FIGURES	**VIII**
	LIST OF TABLES	**IX**
1	**INTRODUCTION**	**01**
	1.1 DIABETES MELLITUS	01
	1.2 RSTUDIO	01
	1.3 OBJECTIVES	04
2	**REVIEW OF LITERATURE**	**05**
	2.1 OVERVIEW OF DIABETES	05
	2.2 GENE EXPRESSION ANALYSIS	05
	2.3 NETWORK BIOLOGY AND HUB GENE IDENTIFICATION	05
	2.4 FUNCTIONAL ENRICHMENT ANALYSIS	05
	2.5 DRUG SCREENING	06
	2.6 MOLECULAR DOCKING AND VISUALIZATION TOOLS	06
3	**METHODOLOGY**	**07**
	3.1 EVALUATION OF DATASET	07
	3.1.1 Pre –processing the data	07
	3.1.2 Filtering the data	07
	3.1.3 Creating an expression object for further analysis	07
	3.2 DEG ANALYSIS	07
	3.2.1 Visualization of DEGs	08
	3.3 DOWNSTREAM ANALYSIS	08
	3.3.1 STRING database	08
	3.3.2 Cytoscape	09
	3.3.3 DAVID database	10
	3.4 MOLECULAR DOCKING	11

3.4.1 Ligand preparation 11
3.4.2 ADME analysis 11
3.4.3 Protein preparation 12
3.4.4 Visualization and Interaction analysis 13

4 RESULT AND DISCUSSION 14

4.1 Pre-processing 14
4.2 DEG analysis 14
4.3 Identification of hub genes and network analysis 17
4.3.1 Top 10 genes 18
4.3.2 Functional enrichment analysis using STRING 18
4.4 DAVID analysis(KEGG pathway) 22
4.5 MOLECULAR DOCKING 25
4.5.1 Protein preparation 25
4.5.2 SwissADME analysis 26
4.5.3 Docking result 26
4.5.4 Protein- ligand visualization 27

5 CONCLUSION 29

6 REFERENCES 30

ABSTRACT

Diabetes mellitus, a chronic metabolic disorder, poses a significant global health challenge, necessitating the discovery of novel therapeutic agents. This study focuses on the identification of potential anti-diabetic phyto-compounds through computational approaches, integrating R programming and molecular docking techniques. Phyto-compounds from natural sources are known for their therapeutic properties, making them ideal candidates for drug discovery.

We employed R for data pre-processing, analysis, and selection of phyto-compounds with known anti-diabetic potential from public databases. The molecular docking simulations were performed using AutoDock Vina. The binding affinities of these phyto-compounds were analysed to assess their potential as inhibitors of these proteins. Our results identified several promising phyto-compounds with strong binding affinities, indicating potential efficacy in managing diabetes mellitus. This study highlights the synergy between bioinformatics tools like R and molecular docking for accelerating the drug discovery process and offers a pathway for developing novel phyto-compound-based therapeutics for diabetes management.

LIST OF FIGURES

FIGURE NO	NAME OF THE FIGURE	PAGE NO
3.1	Steps involved in downstream analysis	08
3.2	String homepage	09
3.3	Steps involved in DAVID	11
4.1	DEG analysis	14
4.2	Volcano plot	15
4.3	Heat map	16
4.4	MA plot	16
4.5	PPI network analysis	17
4.6	PPI network of top 10 genes	18
4.7	String network betweeness of top genes	19
4.8	String network bottleneck of top genes	19
4.9	String network closeness of top genes	19
4.10	String network clustering coefficient of top genes	19
4.11	String network degree of top genes	19
4.12	String network DMNC of top genes	19
4.13	String network eccentricity of top genes	20
4.14	String network EPC of top genes	20
4.15	String network MCC of top genes	20
4.16	String network MNC of top genes	20
4.17	String network radiality of top genes	20
4.18	String network stress of top genes	20
4.19	Pathway enrichment analysis	23
4.20	Insulin resistant pathway	24
4.21	Protein preparation using Autodock vina	25
4.22	Docking score 6NJS with ligand	26
4.23	Protein binding with ligand visualized using PyMol	27
4.24	Docking analysis in PyMol	28

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
01	Datasets Used for Differential Gene Expression Analysis	14
02	Selected Target protein Used for Docking Studies	23

CHAPTER 1

INTRODUCTION

1.1 DIABETES MELLITUS

Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels due to insufficient insulin production or impaired insulin utilization. It affects millions globally, leading to significant health complications, including cardiovascular diseases, neuropathy, and kidney failure. The development of effective drugs to manage or cure diabetes is a critical area of research. Within the age group of 20–79 years, India has 74.9 million diabetics in 2021 projected to increase to 124.9 million by 2045. According to IDF, one out of every seven diabetic adults worldwide resides in India, and one in every third household has diabetic patients

Drug discovery for diabetes involves identifying and optimizing compounds that can regulate blood glucose levels, enhance insulin sensitivity, or stimulate insulin production. With the advent of bioinformatics and computational tools, drug discovery has become increasingly efficient, leveraging large datasets to identify novel drug targets.

Steps in Computational Drug Discovery:

1. Finding the target of disease.
2. Screening of compound.
3. Binding affinity analysis.

1.2 RSTUDIO:

RSTUDIO is a tool involving R programming language. It is used in the data analysis to import, access, transform, plot and model data and for the machine learning to make predictions on data. RStudio is a code editor that comes with syntax highlighting, code completion, and debugging tools. This is

where you write your R code directly into, and these features make the coding process mother and more efficient, which becomes more important as code bases grow in complexity.

There are 2 types of functions namely,

1. User defined function
2. Built in function

User defined function: User-defined functions (UDFs) in R, are functions that you create to perform a task or series of tasks that you need to repeat multiple times. For example, result <- userDefinedFunction(arguments) or print(userDefinedFunction(arguments)).

Built in function: Built-in functions are pre-defined functions that come with a programming language. They are part of the language's library and can be used without additional installation or importation. For example, to find the square root of any value use a function sqrt (value).

Commands used in Rprogramming:

- getwd{} – To give the working directory.
- setwd{} – To change the working directory.
- print{} – To print the variable or value.
- install packages(“”) – To install the packages.
- library(“”) – To load the installed packages.
- read – To read the arguments.
- round – To round the values.
- View() – To view the table.
- rbind() – To combine the rows in the table.
- cbind() – To combine hte columns.
- filter – To filter the values i.e, to remove some kind of variations.

- head – To get the top genes.

Packages used in Rstudio:

For Bulk analysis;

- R.utilis
- Rsubread
- Data.table
- RUVseq
- DESeq2
- Enhanced volcano
- Pheatmap
- RcolorBrewer
- Ggplot2
- edgeR
- EDAseq

The steps involved in the process of analysing the dataset are as follows;

- Viewing the data
- Counting the data
- Filtering the data
- Creating an expression object for further analysis
- Normalization
- Viewing the normalized data using various plots like box plot, volcano plot, violin plot, etc…

Before viewing the data, the dataset can be selected either from Gene omnibus or CancerSEA.

1.3 OBJECTIVES

- To identify differentially expressed genes using R programming.
- To perform protein-protein interaction (PPI) network analysis and identify hub genes.
- To conduct functional enrichment analysis to identify key diabetes-related pathways.
- To select a suitable target protein for drug docking.
- To collect phytocompounds and assess their ADME properties using SwissADME.
- To perform molecular docking using AutoDock Vina.

CHAPTER 2

REVIEW OF LITERATURE

2.1 OVERVIEW OF TYPE 2 DIABETES

Type 2 diabetes mellitus (T2DM) is a multifactorial metabolic disorder associated with chronic inflammation, insulin resistance, and hyperglycemia. It is estimated that by 2045, over 700 million people will be affected worldwide (Sun et al., 2022), highlighting the urgent need for more efficient therapeutic interventions.

2.2 GENE EXPRESSION ANALYSIS

Recent advances in bioinformatics and systems biology have enabled the identification of disease-related biomarkers and therapeutic targets through high-throughput gene expression profiling. Public repositories like the Gene Expression Omnibus (GEO) provide a rich source of transcriptomic data for diseases including T2DM (Lonsdale et al., 2020). Differential expression analysis using R packages such as limma and DESeq2 helps identify genes with significant changes in diabetic conditions (Love et al., 2014).

2.3 NETWORK BIOLOGY AND HUB GENE IDENTIFICATION

Protein-protein interaction (PPI) networks are essential tools for understanding the molecular complexity of diseases. Tools such as STRING (Szklarczyk et al., 2023) and Cytoscape (Otasek et al., 2019) are widely used for network construction and hub gene identification. Hub genes, which show a high degree of connectivity, are often central regulators in disease processes and are valuable targets for drug discovery.

2.4 FUNCTIONAL ENRICHMENT AND PATHWAY ANALYSIS

Functional annotation and enrichment tools like DAVID (Sherman et al., 2022) and KEGG (Kanehisa et al., 2023) provide insight into biological processes, molecular functions, and cellular components associated with differentially expressed genes. In the context of T2DM, enriched pathways typically include insulin resistance, cytokine signaling, and glucose metabolism.

2.5 DRUG SCREENING

For drug screening, natural compounds and phytochemicals have gained attention due to their safety and therapeutic potential (Tiwari et al., 2022). SwissADME is a widely used online platform for assessing drug-likeness and pharmacokinetics of small molecules (Daina et al., 2017). It evaluates GI absorption, blood-brain barrier permeability, and cytochrome P450 inhibition.

2.6 MOLECULAR DOCKING AND VISUALIZATION TOOLS

Molecular docking tools such as AutoDock Vina (Eberhardt et al., 2021) enable virtual screening of ligands against protein targets, predicting binding affinity and interaction patterns. Visualization tools like PyMOL or Discovery Studio assist in analyzing hydrogen bonds and hydrophobic contacts between ligand and protein residues.

CHAPTER 3

METHODOLOGY

3.1 EVALUATION OF DATASET:

By evaluating sample data taken from GEO (https://www.ncbi.nlm.nih.gov/geo/), Diabetes genes were identified using differentially expressed gene analysis in R-studio programming. Using R-studio, we assessed the quality of the genes and normalized the data.

3.1.1 Pre-processing the data:

Pre-processing is the step to view the data, and if the ensemble ID is present, change the ensemble ID into a gene name. There are 2 types of file formats: space-separated file and tab-separated file**.** To view the dataset, the function "View" is used.

3.1.2 Filtering the data:

Performing initial filtering is to remove low-quality genes and cells. Filtering involves the identification of mitochondrial cells. High mitochondrial gene expression can indicate stressed or dying cells.

3.1.3 Creating an expression object for further analysis:

This step involves the creation of set object. Firstly, separate the vectors then combine the vectors and change into a factor. Finally, create an object using the factor and view the set expression set.

3.2 DEG ANALYSIS

Preventing the domination of genes with high expression. Scaling is same as the normalization process. This process involves Technical variability, removing poor quality, improves data comparability. Differentially expressed analysis shows that the genes with significant changes. DESeq2 is a package

used in R studio for DEG analysis. The Normalized genes are compared between normal and disease at logarithm of base 2 of the fold change value and adjusted p-value. Up-regulated and down-regulated genes were obtained based on log2 fold change values. Generally, the functional category criteria were established at an adjusted p-value< 0.05 for significant variations and log2FC> 2 for up-regulated genes. Several visualizing plots such as Dispersion plot, Heat-map and Volcano plot are obtained.

3.2.1 Visualization of DEGs

- **Volcano Plot:** Highlights genes with significant log2 fold change and p-values. Red dots indicate highly significant genes, supporting robust DEG identification.
- **Heatmap:** Clearly distinguishes between control and disease samples based on expression profiles. Clustering patterns indicate biologically relevant segregation.
- **Dispersion Plot:** Displays variability in gene expression data, confirming the presence of both upregulated and downregulated genes

3.3 DOWNSTREAM ANALYSIS:

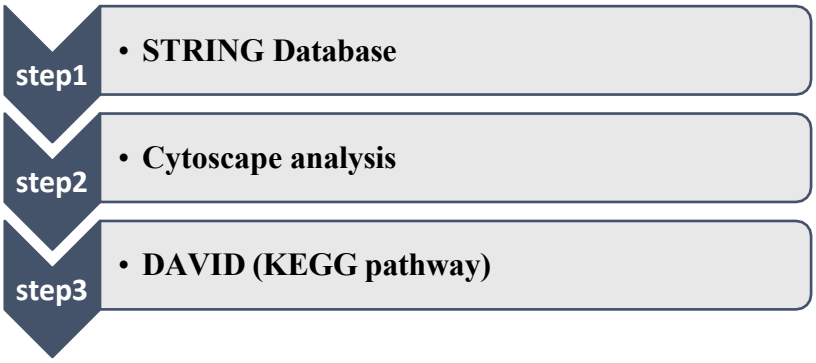

Fig 3.1 Steps involved in downstream analysis

3.3.1 STRING database:

The DEGs were submitted to the STRING database to construct a PPI network. STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) is typically used early in the downstream analysis pipeline. It's a database of known and predicted protein-protein interactions (PPIs). STRING is a biological database used to explore and predict protein-protein interactions (PPIs). It provides a comprehensive resource that integrates known and predicted interactions from various sources, including experimental data, computational prediction methods, and text mining of scientific literature. STRING focuses on understanding the functional links between proteins, which is crucial for various biological processes and disease mechanisms. You might use STRING to:

- Identify potential interactions between your genes or proteins of interest
- Enrich your understanding of biological pathways and processes
- Generate hypotheses for further investigation

By understanding how proteins interact, STRING can help in identifying potential drug targets, particularly in complex diseases where multiple proteins may be involved.

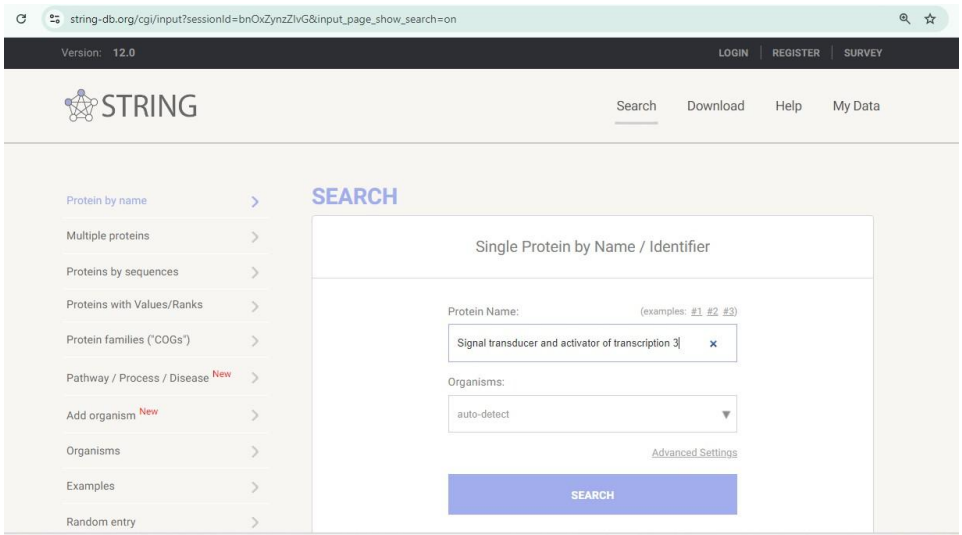

FIG 3.2 Homepage of STRING

3.3.2 Cytoscape:

Cytoscape is a powerful platform for visualizing and analyzing biological networks. It's often used later in the pipeline, after you've gathered more information about your genes or proteins of interest.. The resulting STRING PPI network was visualized using Cytoscape. The CytoHubba plugin in Cytoscape was used to identify the top 10 hub genes based on centrality measures such as Degree, MCC, and Betweenness etc., . Cytoscape used to:

- Visualize and explore complex biological networks
- Integrate data from multiple sources (e.g., gene expression, PPIs,)
- Identify key nodes, hubs, and clusters in your network

3.3.3 DAVID Database:

The DAVID (Database for Annotation, Visualization, and Integrated Discovery) database is commonly used for downstream analysis, particularly after high-throughput experiments like RNA-seq, microarray, or proteomics. The goal of downstream analysis is to interpret the biological meaning behind a list of genes or proteins by exploring their functional roles, pathways, and associations. DAVID is used for

- **Functional Enrichment Analysis**: DAVID identifies which biological functions or pathways are statistically overrepresented in a gene list compared to a background set.
- **Cluster Analysis**: DAVID groups genes into functionally related clusters, providing an overview of how genes may work together in biological processes. These clusters are based on shared annotations, such as involvement in the same pathway or cellular process.

- **Pathway Mapping**: The database can map genes to known biological pathways, helping researchers identify which signalling or metabolic pathways are involved in a particular condition or disease

Steps for Using DAVID for Downstream Analysis:

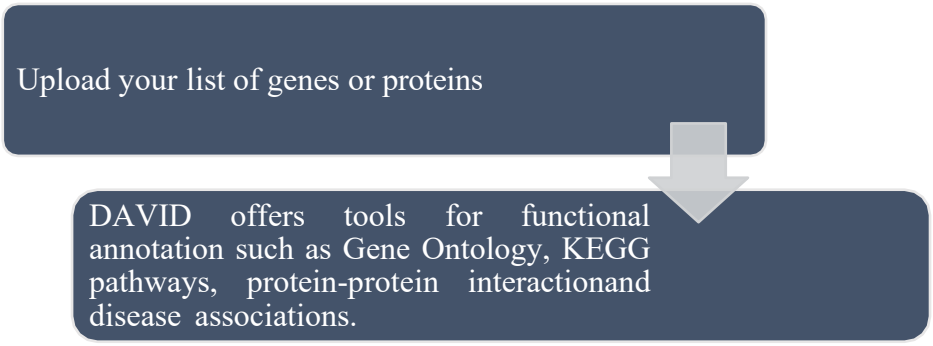

FIG 3.3 Steps involved in DAVID database

3.4 MOLECULAR DOCKING:

3.4.1 Ligand preparation:

Plants which are supposedly having anti-diabetic activity were selected based on their availability and their common use against Diabetes mellitus. Phytocompounds with reported antidiabetic activity were collected from literature, IMPPAT database, and PubChem. Canonical SMILES for 1238 compounds were obtained and submitted to SwissADME.

Total compounds collected: 1238

3.4.2 ADME Analysis (SwissADME)

The pharmacokinetic screening of the selected compounds was conducted using the SwissADME tool. This analysis focused on evaluating drug-likeness properties, gastrointestinal (GI) absorption, blood-brain barrier (BBB) permeability, and interactions with cytochrome P450 enzymes (CYPs).

- **Lipinski's Rule of Five**: Compounds were filtered for drug-likeness based on Lipinski's criteria, including molecular weight, hydrogen bond donors and acceptors, and logP values.
- **Gastrointestinal Absorption**: A majority of the selected compounds showed high GI absorption, indicating their potential for oral bioavailability.
- **BBB Permeability**: Most compounds were not BBB permeant, making them more suitable for peripheral therapeutic action rather than central nervous system effects.
- **Cytochrome P450 Inhibition**: The compounds did not inhibit major CYP enzymes (CYP1A2, CYP2C19, CYP2C9, CYP2D6, and CYP3A4), reducing the risk of drug-drug interactions and suggesting good metabolic stability.

3.4.3 Protein preparation:

From KEGG pathway and string analysis, one gene is selected with highly expressed and linked with Diabetes Mellitus. For the purpose to facilitate docking, the PDB entry was chosen according to its resolution; structures with a resolution of 2.5 Å or higher were preferred as they guaranteed precise atomic positioning. The protein's backbone dihedral angles were also evaluated using Ramachandran plot analysis, where the acceptable threshold for outliers was established at less than 5% to guarantee correct geometric conformations. Every mutation found in the PDB structure was carefully analyzed; if it was determined that a mutation was not indicative of the original biological form, the structure was either rectified or the wild-type form was substituted. After verification, the structure was cleared of any water molecules and unnecessary ligands that would have affected the precision of the docking process. In order to guarantee correct valence and optimize the molecular shape, hydrogen atoms were included into the structure. (PDB ID- 6NJS)

The selected ligands were converted into 3D structures and energy-minimized. Docking was performed using AutoDock Vina. The docking grid was set around the active site of STAT3. Binding affinities were calculated, and the best poses were selected for further visualization. The grid box size was adjusted with AutoDock Tools, a graphical user interface for Auto Dock, so that they encompassed the protein's whole possible binding pocket. The grid box was placed exactly over the active site, and its dimensions (length, width, and height) as well as its center coordinates were carefully adjusted to guarantee that the entire interaction region was covered. The binding pocket size and the expected spectrum of ligand interactions were taken into consideration while determining the grid box's dimensions.

3.4.4 Visualization and Interaction Analysis

After molecular docking, the protein-ligand complexes were visualized using PyMOL to assess the nature of interactions. The selected protein structure was examined for hydrogen bonding, hydrophobic interactions, and spatial fit of the ligand within the active site. Key residues involved in the interactions were highlighted. The visualization confirmed the stable binding of the ligand within the binding pocket, supporting the docking results and validating the ligand’s potential binding efficiency.

CHAPTER 4

RESULT AND DISCUSSION

4.1 PRE-PROCESSING:

For this analysis, we combined the datasets GSE164416, GSE228662 and GSE198922, which were performed as a single project. The raw data obtained from GEO are further assessed to evaluate the quality of genes. After combining datasets, we extracted 24,159 common genes from the above datasets. The combined dataset contains 74 control samples and 180 diseased samples were chosen for further analysis.

Table 4.1 Datasets Used for Differential Gene Expression Analysis

S.NO	ACCESSION ID	CONTROL	DISEASED
1	GSE164416	16	41
2	GSE228662	0	67
3	GSE198922	56	74
	PROCESSED SAMPLES	74	180

4.2 DEG ANALYSIS:

Genes with adjusted p-value < 0.05 and absolute log2 fold change > 1 were considered differentially expressed. This resulted in the identification of 7,564 DEGs. Data visualization was performed using volcano plots, heatmaps, and dispersion plots.

Fig 4.1 DEG analysis

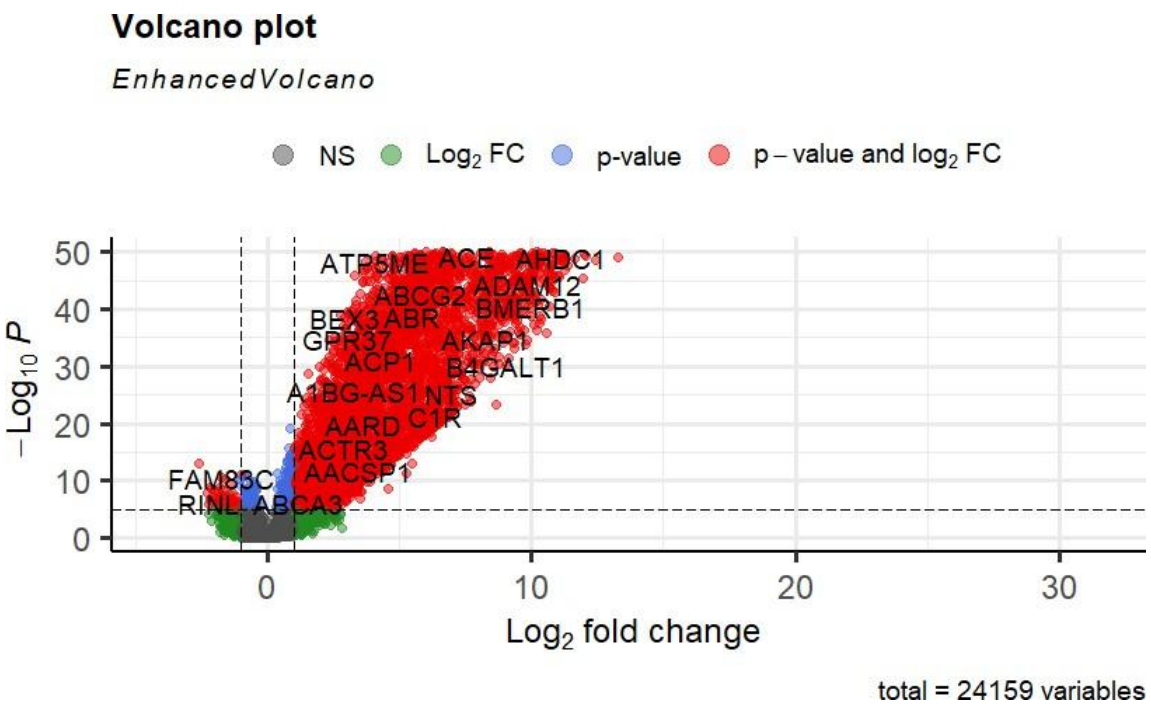

Fig 4.2 Volcano plot showing DEG distribution

This plot represents Volcano plot of highly expressed genes in the samples. This shows the values of -log10 P and log2 foldchange. Red color denotes the upregulated genes whereas green color denotes down-regulated genes. The default line can be adjusted according to the p-value. Total variable count is 24159.

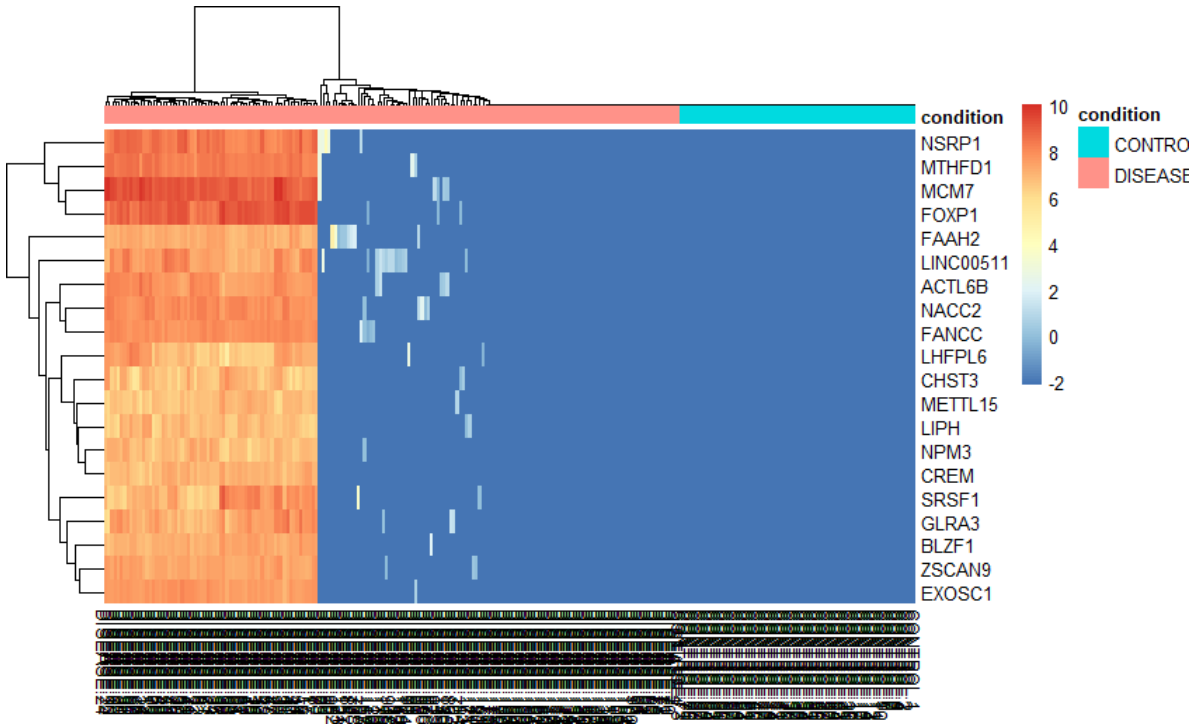

Fig-4.3 Heatmap

Heatmap depicts the variance of differentially expressed genes in different samples. Diseased genes are denoted by dark orange to red color while control or healthy genes denoted as blue. Clustering patterns indicate biologically relevant segregation.

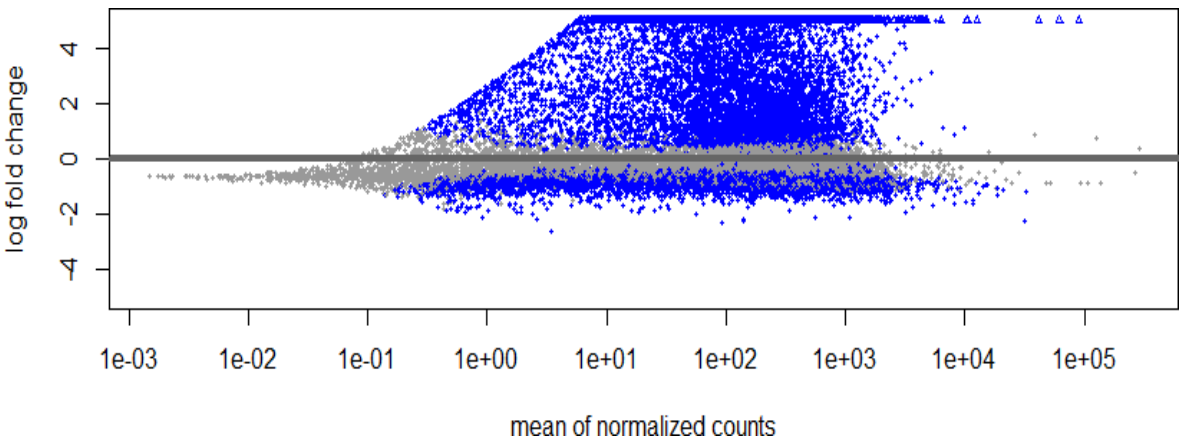

Fig 4.4 MA Plot

MA plot can be used to display a difference in expression patterns between samples. On the Y-axis, it shows the difference in expression of the clusters and on the X-axis, it shows the average contribution of the clusters. Each dot is a gene, and the genes that show the most differential expression are highlighted in blue and labelled. The rest are grey and unnamed.

4.3 IDENTIFICATION OF HUB GENES AND NETWORK ANALYSIS:

The network analysis and interaction between the proteins for differentially expressed genes were established using Cytoscape software. Cytoscape is an open-source software platform for visualizing molecular interaction networks. Cytohubba were installed in Cytoscape for hub gene analysis. The differentially expressed genes were imported in Cytoscape, the 12 type of networks analysis such as MCC, MNC, DMNC, EPC, BottleNeck, EcCentricity, Closeness, Radiality, Betweeness, Stress and ClusteringCoefficient were done. By using these network analysis, we were identified the top 10 DEG associated with hub genes.

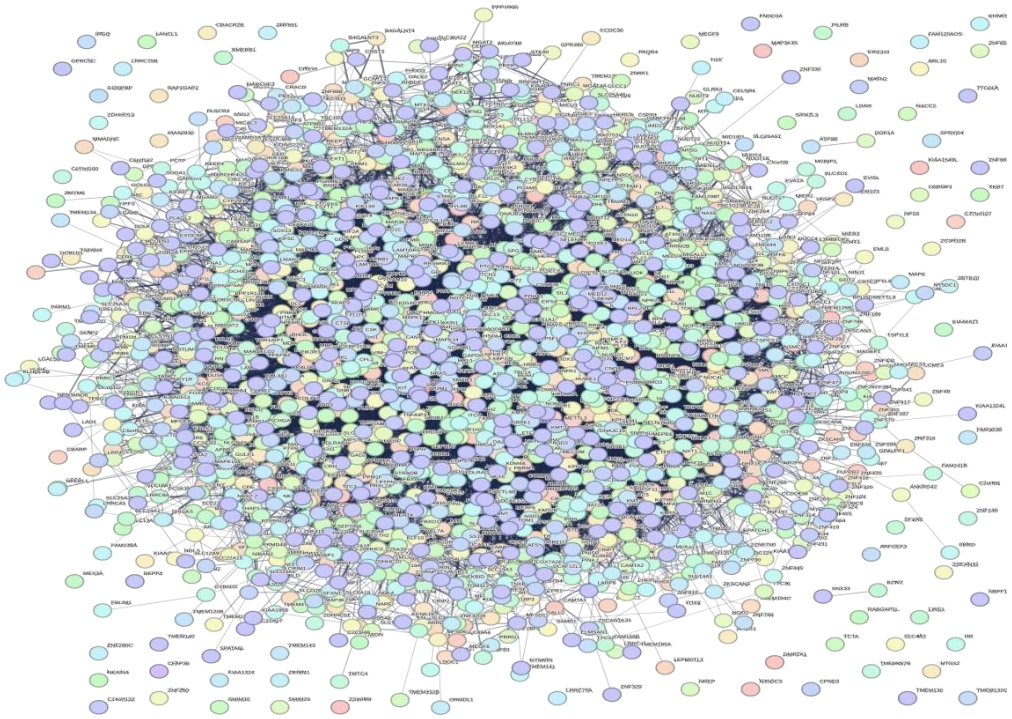

Fig 4.5 Protein-Protein interactions using String

4.3.1 Top 10 genes:

STAT3

AKT1

CTNNB1

ACTB

GAPDH

HSP90AA1

HSPA8

TRIM28

RPS27A

HSP90AB

The occurrence of filtered genes in all the 12 methods of CytoHubba was depicted in below figures.

4.3.2 Functional Enrichment Analysis of T2DM using STRING

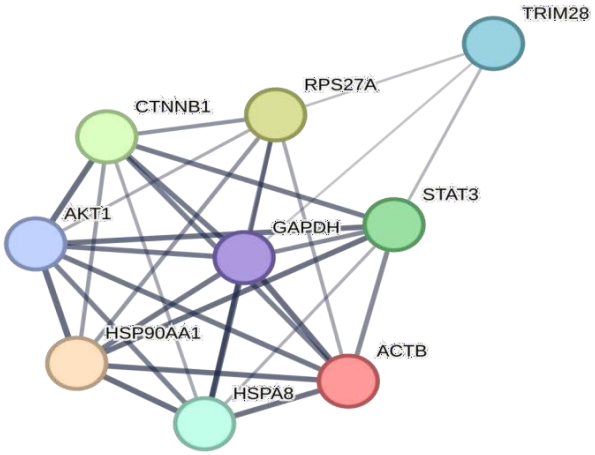

Fig 4.5 PPI interaction of Top 10 genes

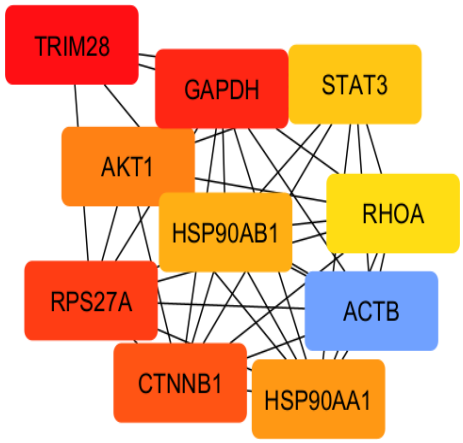

Fig 4.7 Betweeness

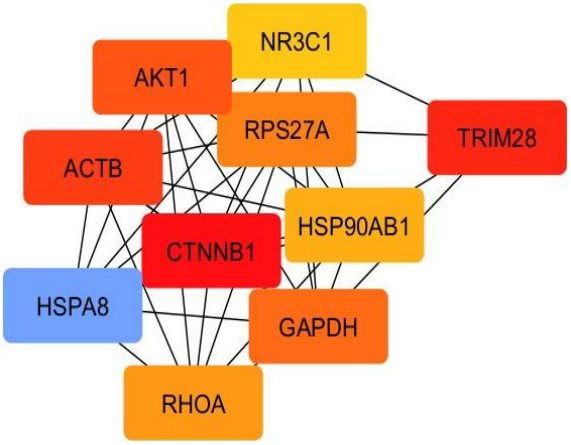

Fig 4.8 Bottleneck

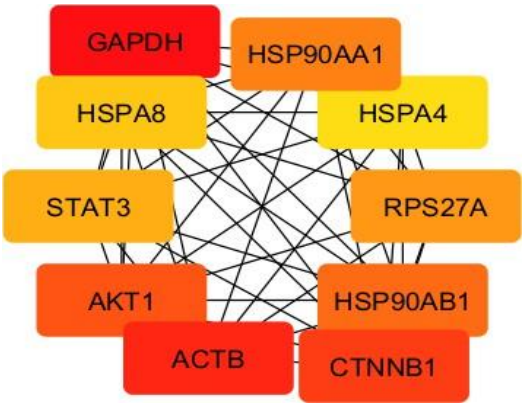

Fig 4.9 Closeness

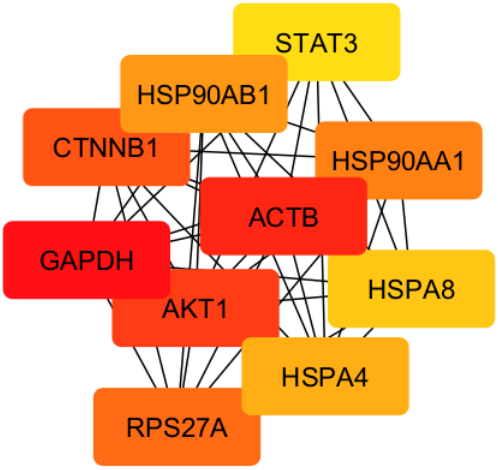

Fig 4.10 Degree

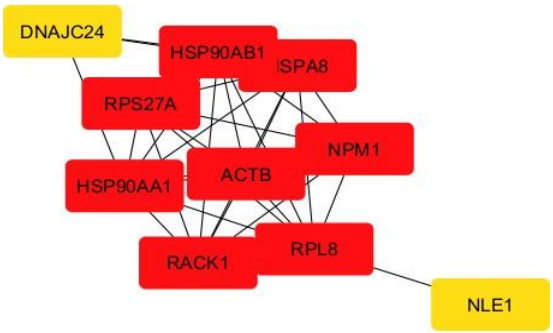

Fig 4.11 EcCentricity

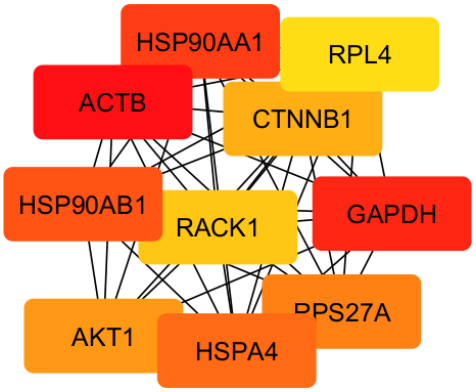

Fig 4.12 EPC

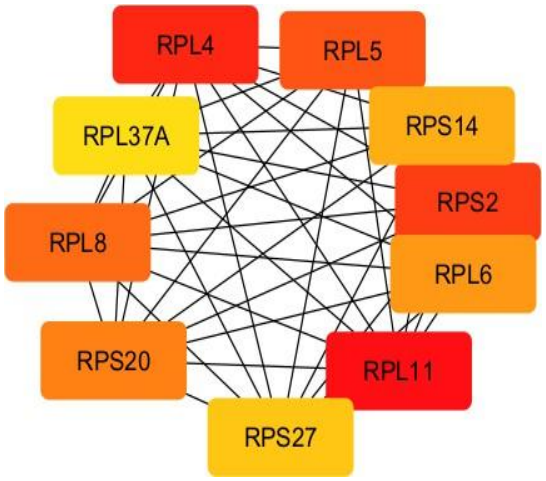

Fig 4.13 MCC

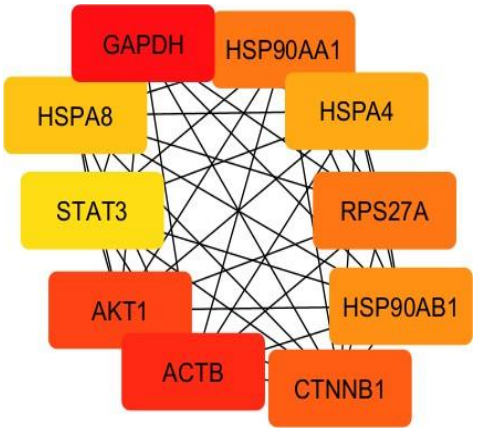

Fig 4.14 MNC

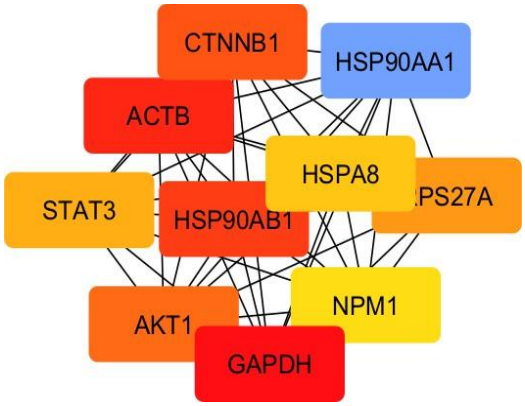

Fig 4.15 Radiality

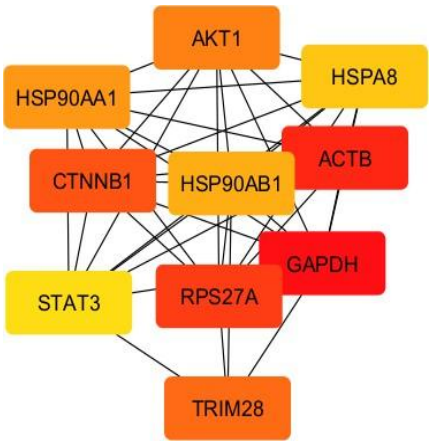

Fig 4.16 Stress

Fig 4.17 Clustering co-efficient

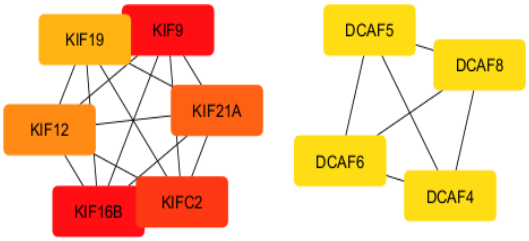

Fig 4.18 DMNC

4.4 DAVID ANALYSIS (KEGG PATHWAY):

DAVID was used to perform gene ontology (GO) and KEGG pathway enrichment analysis. The KEGG pathway enrichment analysis using DAVID identified **insulin resistance** as one of the significantly enriched pathways, with a p-value of 7.6e-5 and an FDR of 0.001**,** indicating a strong association of the differentially expressed genes with this pathway.

DAVID: Database for Annotation, Visualization, and Integrated Discovery (Laboratory of Human Retrovirology and Immunoinformatics (LHRI); National Institute of Allergies and Infectious Diseases (NIAID); Leidos Biomedical Res...

davidbioinformatics.nih.gov/chartReport.jsp?annot=52

KEGG_PATHWAY	Salmonella infection	RT	128	1.7	1.7E-5	4.4E-4
KEGG_PATHWAY	Spinocerebellar ataxia	RT	80	1.1	1.9E-5	4.5E-4
KEGG_PATHWAY	Shigellosis	RT	127	1.7	2.3E-5	5.0E-4
KEGG_PATHWAY	Regulation of actin cytoskeleton	RT	119	1.6	2.5E-5	5.0E-4
KEGG_PATHWAY	Parkinson disease	RT	136	1.8	2.5E-5	5.0E-4
KEGG_PATHWAY	Sphingolipid signaling pathway	RT	69	0.9	3.6E-5	6.7E-4
KEGG_PATHWAY	Endocytosis	RT	127	1.7	3.7E-5	6.7E-4
KEGG_PATHWAY	Axon guidance	RT	97	1.3	3.9E-5	6.7E-4
KEGG_PATHWAY	Adherens junction	RT	55	0.7	5.0E-5	8.2E-4
KEGG_PATHWAY	MAPK signaling pathway	RT	147	1.9	5.4E-5	8.5E-4
KEGG_PATHWAY	Pancreatic cancer	RT	47	0.6	6.8E-5	1.0E-3
KEGG_PATHWAY	Cellular senescence	RT	84	1.1	7.0E-5	1.0E-3
KEGG_PATHWAY	Insulin resistance	RT	62	0.8	7.6E-5	1.0E-3
KEGG_PATHWAY	Lysosome	RT	73	1.0	7.8E-5	1.0E-3
KEGG_PATHWAY	N-Glycan biosynthesis	RT	35	0.5	1.4E-4	1.8E-3
KEGG_PATHWAY	Nucleocytoplasmic transport	RT	60	0.8	2.4E-4	3.0E-3
KEGG_PATHWAY	Colorectal cancer	RT	50	0.7	3.1E-4	3.7E-3
KEGG_PATHWAY	Galactose metabolism	RT	23	0.3	3.6E-4	4.0E-3
KEGG_PATHWAY	Pathways of neurodegeneration - multiple diseases	RT	219	2.9	3.6E-4	4.0E-3

Fig 4.19 Pathway enrichment analysis

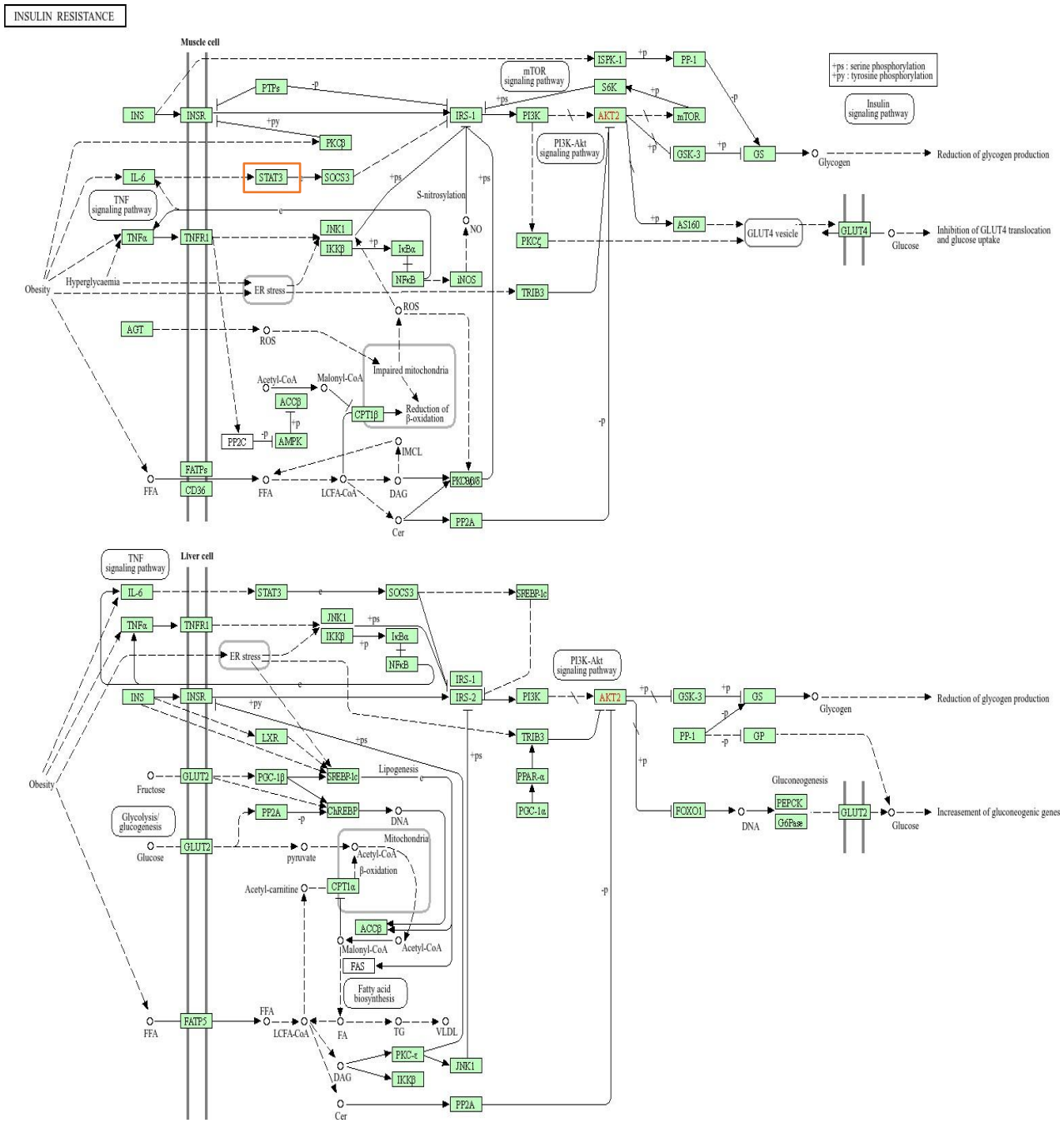

FIG 4.20 Insulin Resistance pathway

KEGG pathway analysis of insulin resistance pathway in which **STAT3** gene involved. STAT3 gets activated by inflammatory molecules like TNF-α and IL-6 (these increase in diabetes). Once activated, STAT3 increases another protein called SOCS3, which blocks IRS1- a key step in the insulin signaling pathway.

4.5 MOLECULAR DOCKING

4.5.1 Protein preparation

Gene	PDB ID	UniProt ID	Protein name
STAT3	6NJS	P40763	Signal transducer and activator of transcription 3

Table 4.2 Selected Target protein Used for Docking Studies

The protein structure used for docking was retrieved from the Protein Data Bank in PDB format. Before proceeding to docking, the structure was preprocessed using AutoDock Tools. This involved the removal of crystallographic water molecules, which can interfere with ligand binding if not functionally relevant. Polar hydrogens were then added to the structure to simulate realistic hydrogen bonding during docking. Kollman charges were also assigned to the protein. The preprocessed protein structure was saved in the PDBQT format, which is required for docking in AutoDock Vina. This preparation step ensured that the protein was in the optimal form for accurate ligand docking and interaction analysis.

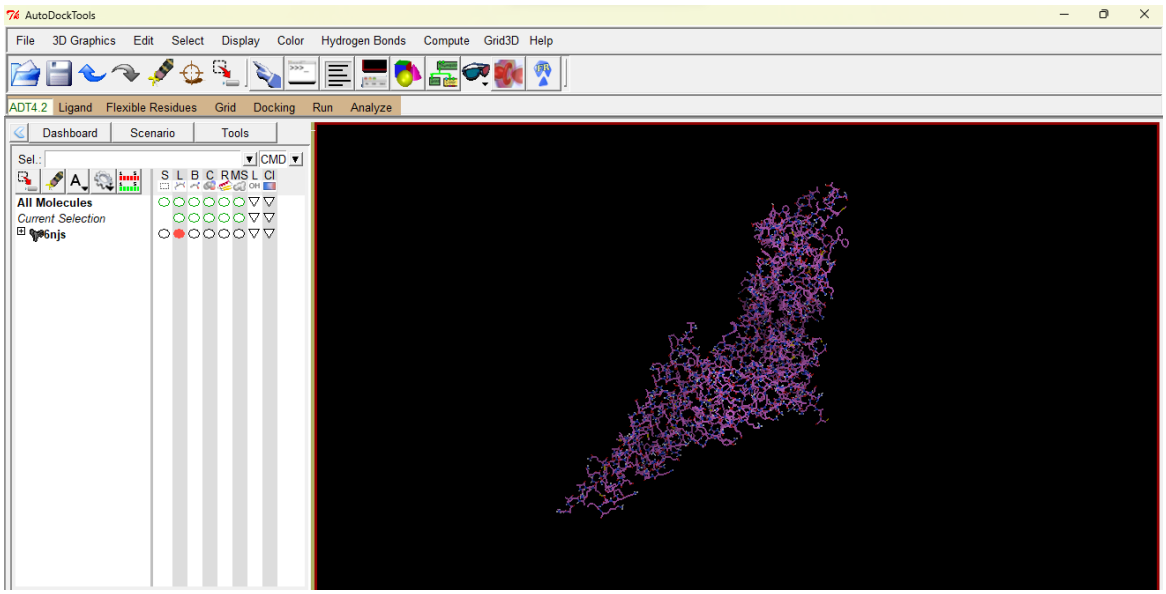

FIG 4.21 Protein preparation using Autodock vina (PDB ID: 6NJS)

4.5.2 Swiss ADME analysis

SwissADME analysis helped shortlist 44 compounds with favorable ADME profiles, which were considered suitable candidates for molecular docking studies.

4.5.3 Docking result

- **Tool Used:** AutoDock Vina
- **Target Protein:** STAT3 (6NJS)
- **Top Ligand:** Leucocyanidin
- **Binding Affinity:** –8.1 kcal/mol

The molecular docking results revealed that the selected ligand showed strong binding affinity to the target protein's active site. A binding energy of -8.1 kcal/mol indicates high stability and strong interaction, as more negative values represent better binding.

```
WARNING: The search space volume > 27000 Angstrom^3 (See FAQ)
Output will be lig14_out.pdbqt
Detected 2 CPUs
Reading input ... done.
Setting up the scoring function ... done.
Analyzing the binding site ... done.
Using random seed: 1363622024
Performing search ...
0%   10   20   30   40   50   60   70   80   90   100%
|----|----|----|----|----|----|----|----|----|----|
***************************************************
done.
Refining results ... done.

mode |   affinity | dist from best mode
     | (kcal/mol) | rmsd l.b.| rmsd u.b.
-----+------------+----------+----------
   1         -8.1      0.000      0.000
   2         -8.1      1.341      6.650
   3         -7.4     26.012     27.230
   4         -7.4     34.611     36.317
   5         -7.3     22.264     23.568
   6         -7.2     22.926     24.574
   7         -7.2     25.863     27.097
   8         -7.1     22.803     24.654
   9         -7.0     26.309     27.144
```

Fig 4.22 Docking score of 6NJS protein with leucocyanidin(Lig.14)

4.5.4 Protein-Ligand Visualization

To further understand the binding interaction, the complete protein-ligand complex was visualized using PyMOL. The ligand was observed to occupy a defined pocket within the protein structure, surrounded by amino acid residues that stabilize the complex.

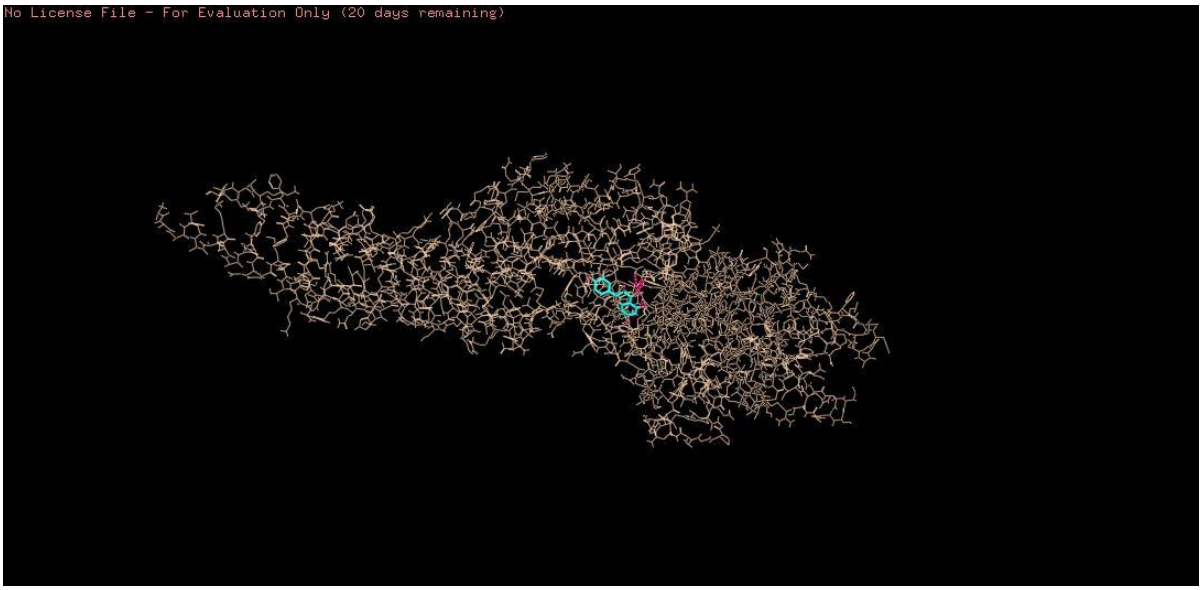

Fig 4.23 Protein binding with ligand visualized using PyMOL

The entire protein structure (beige lines) with a ligand molecule (cyan) bound inside the binding pocket. The protein is shown in wireframe style, representing its 3D conformation. The small colored structure at the center is the docked ligand. The location of the ligand indicates that it has bound deep inside the active site, suggesting good spatial compatibility and docking accuracy.

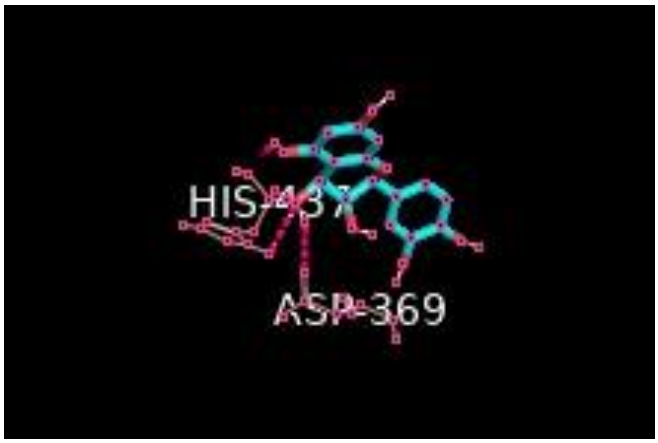

FIG-4.24 Docking analysis in PyMOL

A focused view highlighted two key interacting residues,

- HIS-437 (Histidine 437)
- ASP-369 (Aspartic acid 369**)**

These are key residues in the protein's active site interacting with the ligand. These residues form polar interactions with the ligand, suggesting strong and specific binding within the active site. The geometry and orientation of the ligand indicate optimal docking pose, reinforcing the accuracy of the AutoDock Vina predictions.

CHAPTER 7

CONCLUSION

An integrated computational workflow enabled the identification of a diabetes-associated protein target along with potential plant-based ligands. Differential gene expression analysis revealed biologically significant genes related to type 2 diabetes, with network analysis pointing to central regulators involved in insulin resistance. Enrichment results confirmed the relevance of these genes in key signaling pathways, particularly those influencing glucose metabolism and inflammatory response.

Subsequent pharmacokinetic screening ensured selection of ligands with desirable absorption, safety, and drug-likeness profiles. Molecular docking demonstrated strong and stable interactions between the lead compound, **Leucocyanidin**, and active site residues of the target protein. Leucocyanidin, a flavonoid compound derived from *Ficus benghalensis* (Indian banyan tree), exhibited promising potential as a novel antidiabetic agent. The findings provide a valuable direction for the development of therapeutic candidates through rational design and further experimental validation

REFERENCES:

1. Bayarsaikhan, B., Zsidó, B. Z., Börzsei, R., & Hetényi, C. (2024). Efficient Refinement of Complex Structures of Flexible Histone Peptides Using Post-Docking Molecular Dynamics Protocols. *International Journal of Molecular Sciences*, *25*(11), 5945.
2. Chen, M., Zhu, Y., Li, H., Zhang, Y., & Han, M. (2022). A quantitative proteomic approach explores the possible mechanisms by which the small molecule stemazole promotes the survival of human neural stem cells. *Brain sciences*, *12*(6), 690.
3. Prottoy, N. I., Sarkar, B., Ullah, A., Hossain, S., Boby, A. S., & Araf, Y. (2019). Molecular docking and pharmacological property analysis of antidiabetic agents from medicinal plants of Bangladesh against type II diabetes: a computational approach. *PharmaTutor*, *7*(9), 6-15.
4. Natarajan, A., Sugumar, S., Bitragunta, S., & Balasubramanyan, N. (2015). Molecular docking studies of (4 Z, 12 Z)-cyclopentadeca-4, 12-dienone from Grewia hirsuta with some targets related to type 2 diabetes. *BMC complementary and alternative medicine*, *15*, 1-8.
5. El-Sherbiny, M., El-Sayed, R. M., Helal, M. A., Ibrahiem, A. T., Elmahdi, H. S., Eladl, M. A., ... & Zaitone, S. A. (2021). Nifuroxazide mitigates angiogenesis in ehlrich's solid carcinoma: molecular docking, bioinformatic and experimental studies on inhibition of il-6/jak2/stat3 signaling. *Molecules*, *26*(22), 6858.
6. Dwivedi, P. S., Patil, R., Khanal, P., Gurav, N. S., Murade, V. D., Hase, D. P., ... & Gurav, S. S. (2021). Exploring the therapeutic mechanisms of Cassia glauca in diabetes mellitus through network pharmacology, molecular docking and molecular dynamics. *RSC advances*, *11*(62), 39362-39375.

7. Singh, P., Singh, V. K., & Singh, A. K. (2019). Molecular docking analysis of candidate compoundsderived from medicinal plants with type 2 diabetes mellitus targets. *Bioinformation*, *15*(3), 179.
8. Prieto-Martínez, F. D., & Medina-Franco, J. L. (2021). Flavonoids as putative epi-modulators: Insight into their binding mode with BRD4 bromodomains using molecular docking and dynamics. *Frontiers in Chemistry*, 9, 678234.
9. Singh, P., Singh, V. K., & Singh, A. K. (2022). Molecular docking analysis of candidate compounds derived from medicinal plants with type 2 diabetes mellitus targets. *Journal of Biomolecular Structure and Dynamics*, 40(3), 1023–1035.
10. James, P., Davis, S. P., Ravisankar, V., Nazeem, P. A., & Mathew, D. (2021). Novel antidiabetic molecules from the medicinal plants of Western Ghats of India, identified through wide-spectrum in silico analyses. *Molecular Diversity*, 25, 847–860.
11. Patil, V. S., Deshpande, S. H., Harish, D. R., Patil, A. S., Virge, R., Nandy, S., & Roy, S. (2020). Gene set enrichment analysis, network pharmacology and in silico docking approach to understand the molecular mechanism of traditional medicines for the treatment of diabetes mellitus. *Journal of Ethnopharmacology*, 256, 112775.
12. Bayarsaikhan, B., Zsidó, B. Z., Börzsei, R., & Hetényi, C. (2020). Efficient refinement of complex structures of flexible histone peptides using post-docking molecular dynamics protocols. *Journal of Chemical Information and Modeling*, 60(11), 5520–5533.
13. Prottoy, M. N. I., Ullah, M. A., Sarkar, B., Hossain, S., Boby, A. S., & Araf, Y. (2022). Molecular docking and pharmacological property analysis of antidiabetic agents from medicinal plants of Bangladesh against type II diabetes: A computational approach. *BioMed Research International*, 2022, 8710983.

14. Peng, Y. L., Zhang, Y., Pang, L., Dong, Y. F., Li, M. Y., Liao, H., & Li, R. S. (2023). Integrated analysis of single-cell RNA-Seq and bulk RNA-Seq combined with multiple machine learning identified a novel immune signature in diabetic nephropathy. *Frontiers in Endocrinology*, 14, 1133920. https://doi.org/10.3389/fendo.2023.1133920
15. Daina, A., Michielin, O., & Zoete, V. (2017). SwissADME: A free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Scientific Reports*, 7, 42717.
16. Eberhardt, J., Santos-Martins, D., Tillack, A. F., & Forli, S. (2021). AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings. *Journal of Chemical Information and Modeling*, 61(8), 3891–3898
17. Kanehisa, M., Furumichi, M., Sato, Y., Kawashima, M., & Ishiguro-Watanabe, M. (2023). KEGG for taxonomy-based analysis of pathways and genomes. *Nucleic Acids Research*, 51(D1), D587–D592. https://doi.org/10.1093/nar/gkac963
18. Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 550. https://doi.org/10.1186/s13059-014-0550-8
19. Lonsdale, J., et al. (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509), 1318–1330. https://doi.org/10.1126/science.aaz1776
20. Otasek, D., Morris, J. H., Bouças, J., Pico, A. R., & Demchak, B. (2019). Cytoscape automation: Empowering workflow-based network analysis. *Genome Biology*, 20, 185. https://doi.org/10.1186/s13059-019-1758-4
21. Sherman, B. T., Hao, M., Qiu, J., Jiao, X., Baseler, M. W., Lane, H. C., Imamichi, T., & Chang, W. (2022). DAVID: A web server for functional enrichment analysis and functional annotation of gene lists (2021 update).

Nucleic Acids Research, 50(W1), W216–W221. https://doi.org/10.1093/nar/gkac194

22. Szklarczyk, D., Kirsch, R., Koutrouli, M., Nastou, K. C., Mehryary, F., Hachilif, R., ... & von Mering, C. (2023). The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1), D638–D646. https://doi.org/10.1093/nar/gkac1000
23. Sun, H., Saeedi, P., Karuranga, S., Pinkepank, M., Ogurtsova, K., Duncan, B. B., ... & IDF Diabetes Atlas Study Group. (2022). IDF Diabetes Atlas: Global and regional diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Research and Clinical Practice*, 183, 109119. https://doi.org/10.1016/j.diabres.2021.109119
24. Tiwari, V., Tiwari, R., Sinha, R. A., & Yadav, D. (2022). Phytochemicals as potential therapeutic agents against diabetes: A mechanistic review. *Biomedicine & Pharmacotherapy*, 149, 112623. https://doi.org/10.1016/j.biopha.2022.112623